

STAT8020 Project

Group Members



Department of Statistics and Actuarial Science

The University of Hong Kong

Contents

1. INTRODUCTION	1
2. PROPOSED STRATEGY	2
2.1 RATIONALE	2
2.2 TRADING ALGORITHM	2
3. BACKTESTING	3
3.1 PARAMETER SELECTION	3
3.2 BACK-TESTING RESULT	4
4. OUT-SAMPLE TESTING	5
5. OBSERVATIONS AND DISCUSSIONS	5
5.1 REMOVE MEAN-REVERSION POSITION	6
5.2 ADD TREND SIGNAL	6
6. CONCLUSION	7
7. REFERENCE	8
8. APPENDIX	9

1. Introduction

Efficient Market Hypothesis of semi-strong form states that the price will be reflected in all publicly available information fully and immediately, thus no trading strategy can beat the market portfolio as the market is in equilibrium. However, this traditional model failed to explain many market anomalies which started to accumulate since the 1980s. Short-term momentum and long-term contrarian are two of the market anomalies.

By momentum strategy, traders will take a long (short) position with the hope that the upward (downward) momentum will continue once the stock's price rises (declines). On the other hand, contrarian strategy states that the price will reverse to the mean and long-term returns are negatively correlated. In real situations, the market is not so efficient because of investors' short-term underreaction and long-term overreaction, which indicates that there are arbitrage chances.

The Home Depot, Inc. (HD) is one of the dominant home improvement retailers. The historical data are collected from Yahoo Finance which contains the daily open price, closing price, adjusted price and volume. As shown in Figure 1, the closing price of HD fluctuated in a range between 20 and 40 before entering the bull market and increasing sharply to 120 in 2016. This project would like to apply portfolio theory on strategy diversification to HD stock, and have a portfolio of strategies in which momentum and mean-reversion strategy maintains a different weight.



Figure 1 HD close price

2. Proposed strategy

2.1 Rationale

To define the trend, exponentially weighted moving average (EWMA) is used as tool for momentum strategy.

As a supplement to Bollinger bands, the Z-score indicator is used to assess the position of the price for contrarian strategy. Z-score indicator in this project is defined as the negative Z-score calculated in statistics. A positive value of Z-score indicator indicates the score is below the mean and investors can take a long position based on mean-reversion theory.

Other than portfolio of stocks or bonds, a portfolio of strategy with the combination of momentum and contrarian trading is considered in this project because it's believed that the two may have effects at the same time but with different important weights for different holding periods. Rather than one-time buying or selling, the strategy would give the exact trade position by smoothing.

2.2 Trading Algorithm

Based on the momentum and contrarian trading ideas, this project proposes the following combined strategy.

- 1) Calculate the Z-score indicator as Reversion signal with rolling parameter `N_LAG_REVERSION`.

$$Reversion_t = Zscore_t = - \frac{Return - Mean_{(N_LAG_REVERSION)}}{Sd_{(N_LAG_REVERSION)}}$$

- 2) Calculate the EWMA, define Momentum signal with rolling parameter `DECAY_FACTOR_OF_EWMA`.

$$Momentum_t = P_t - EWMA_{(DECAY_FACTOR_OF_EWMA)}$$

- 3) Normalize the $Reversion_t$ and $Momentum_t$ into $[0,1]$ by min-max scaling.
- 4) Calculate trade position by averaging Momentum and Reversion with rolling parameter `N_LAG_SIGNAL_DECAY`, and renormalize the trade position into $[0,1]$. (The unequal weights for reversion and momentum indicators could be a further improvement approach considering an additional weight indicator to

describe the fact that sometimes reversion will be more significant and sometimes momentum becomes better.)

$$Trade\ position_t = (Reversion_t + Momentum_t)_{(N_LAG_SIGNAL_DECAY)}$$

3. Backtesting

This project constructs a backtesting by using the adjusted closing price of ticker (HD) from 2000 to 2015, As mentioned before, the position used in this project is decided by the combination of Z-score and EWMA as well as its smoothing parameter t. The backtesting methodology is as followed.

3.1 Parameter selection

For the lag of Z-score, this project set it by 5days ,8days, 10days, 20days, 60days, 100days separately, the output is as followed:



Figure 2 Backtesting for rolling parameter in contrarian strategy

For the decay factor of EWMA, set it by 0.05, 0.25, 0.5 ,0.75 ,0.96 ,0.999 separately, the output is as followed:

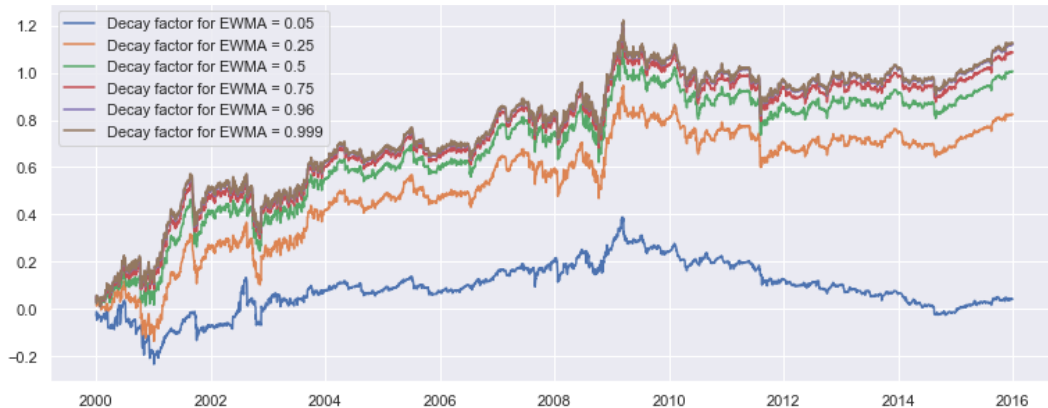


Figure 3 Backtesting for rolling parameter in momentum strategy

For the signal decay (smoothing parameter of position), set it by 2days, 5days, 10days, 20days ,40days separately, the output is as followed:



Figure 4 Backtesting for smoothing parameter in overall strategy

3.2 Back-testing result

It can be found that when setting the lag of Z-score by 5days, 0.999 for decay factor of EWMA, 5days for the signal decay of the position, it can maximize the profit in training dataset. To be mentioned, weighting in EWMA does not seem to be needed, due to “0.999” is close to “1”.

Then compare the performance between the proposed strategy by using the best parameter and buy-and-hold in training dataset. The result is as followed:

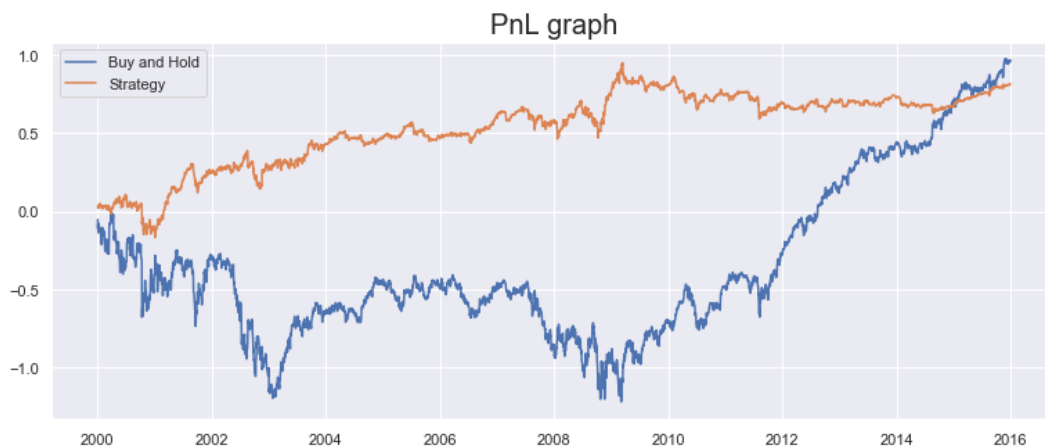


Figure 5 Comparison between proposed strategy and BAH

It can be concluded that in training dataset, by using the best parameter, the proposed strategy performs better than buy-and-hold though it may not always win BAH

especially after 2015. However, it is more stable than buy-and-hold from the long-term trading perspective.

4. Out-Sample Testing

Based on backtesting result, this project set the lag of Z-score by 5 days, 0.999 for decay factor of EWMA, 5 days for the signal decay of our position for the out-sample testing. The result is as followed:

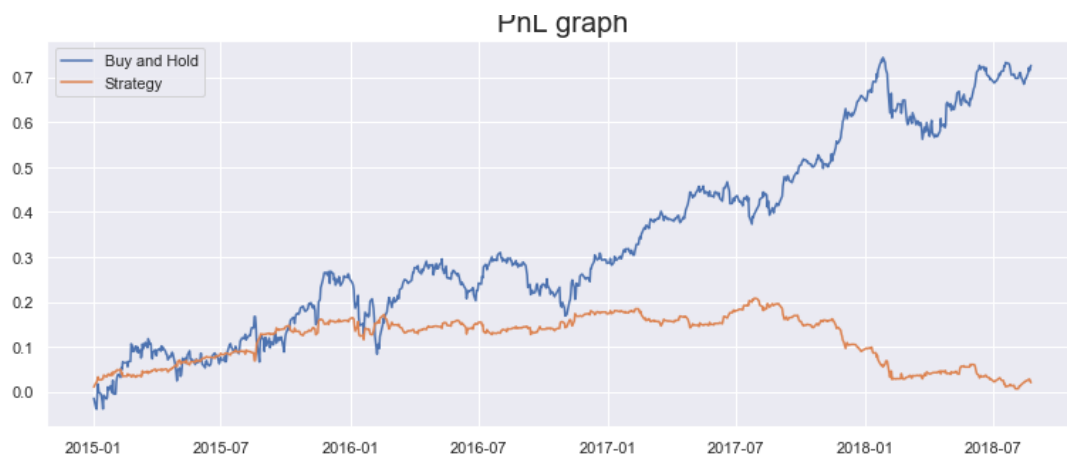


Figure 6 Result for out-sample test

Clearly, the proposed strategy can't beat buy-and-hold strategy in the testing dataset. As what mentioned in introduction, after 2014, the stock price grew rapidly at an exponential rate, and the entire trading market was in a bull market. The strategy proposed in this project replays the weights of momentum and mean-version to be equal, which led to a result that there is no way for the trend of momentum to be better displayed in the trading signals, so that it performs poorly in the bull market outside the training sample.

5. Observations and Discussions

From the above analysis, it can be shown that the strategy composed of the best parameters selected by backtesting does not perform well outside the sample. Returning to the stock itself, the difference in performance is mainly because the upward trend of stock price outside the sample is much higher than that in the sample. Based on this, the essential reasons for the weak performance of the strategy proposed in out-sample

area are explored.

5.1 Remove mean-reversion position

In the proposed strategy, it is mainly composed of two parts, and part of it is that the stock price will return to the mean level strategy (Z-score) when the stock deviates from the mean, and the other part is the momentum strategy (EWMA) that the stock price will continue to a tendency when there is a trend. It can be found that in the bull market, the trend of rising has no way to be sensitively perceived in the strategy, so this may be due to the influence of EWMA. The result of strategy with the Z-score removed is shown in the following figure. Comparing with Figure 6 and Figure 5, it can be seen that before 2014, the performance of the following figure is not as good as Figure 5; and after 2014, its performance is a little better than the first half of Figure 6 (more robustly rising), This confirms that the reason for the weak performance of the strategy is that EWMA can't show its advantages when the trend appears.

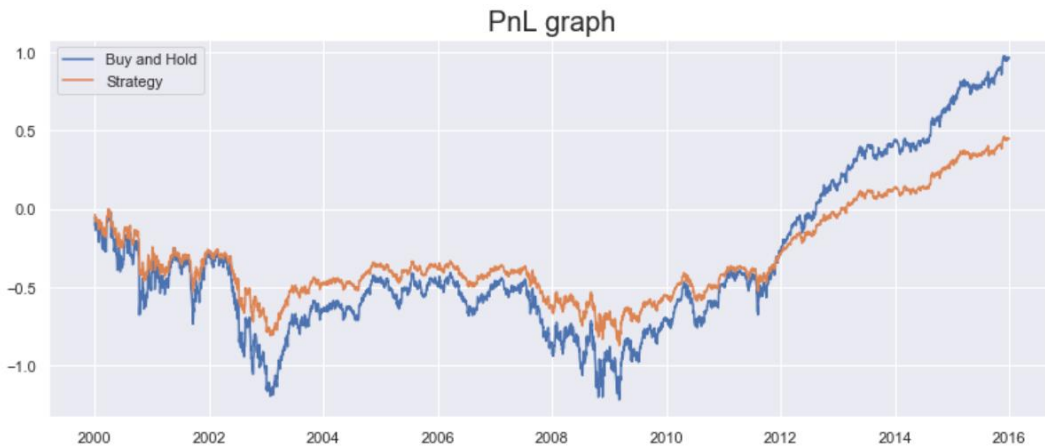


Figure 7 Result for removing position determined by Z-score

5.2 Add trend signal

In summary, Z-score is very effective when the market is down or stable, while Z-score is almost ineffective when the market is going up, and EWMA is relatively effective. Therefore, adding a signal to judge the beginning of the trend will be effective. Improve this strategy and use the momentum strategy when the trend signal appears.

The establishment of the efficient market hypothesis is equivalent to the fact that the

stock price has fully reflected all known information, so any information that affects the price can predict the stock's rate of return (Fama, 1998). In fact, any variable with a coefficient of return that is not zero can predict the trend of the rate of return.

In this regard, the improvement proposed in this project is to add the following trend start signal:

$$\text{Dividend yield} = (\text{Dividend} / \text{Price ratio})$$

In theory, the dividend yield will change as the stock price rises and falls. It is also the ratio that determines the return of the cash flow that investors will receive and is a good indicator of the rise or fall of the stock price (Maio & Santa-Clara, 2015). It is easy to implement a regression between return and dividend yield to find the correlation coefficient. Therefore, as long as the dividend rate changes more than this correlation coefficient, it can be considered that the stock price will have a rising signal to a large extent, and vice versa.

6. Conclusion

The combination of momentum and contrarian strategies presented in this project performed well in the market down or shock and can completely defeat BAH and achieve higher returns. Moreover, when selecting the rolling parameter for Z-score, the smaller the parameter, the more frequent the transaction hidden, the higher the transaction cost, but the optimal parameter is generated at a small level, which indicates the dynamic standardization made a very good smoothing effect in the mean-reversion strategy. On the contrary, when the market is going up, the strategy proposed in this project is not good. This is mainly because Z-score completely loses its advantages and EWMA has no way to show the trend well. Therefore, it is necessary to add trend signals to this strategy for improvement. But overall, the long-term performance of the strategy is still well.

Reference

- Fama, E. F. (1998). Market efficiency, long-term returns, and behavioral finance. *Journal of financial economics*, 49(3), 283-306.
- Maio, P., & Santa-Clara, P. (2015). Dividend yields, dividend growth, and return predictability in the cross section of stocks. *Journal of Financial and Quantitative Analysis*, 50(1-2), 33-60.

Appendix

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
'''
helper functions
'''

def transform_time(_df, time_col_name='', form='%Y%m%d'):
    df = _df.copy()
    from datetime import datetime
    df.loc[:, time_col_name] = df[time_col_name].map(lambda x:
datetime.strptime(str(x), form))
    return df

def get_slice(_df, column_name, to_keep):
    if type(to_keep) == type(lambda x: x):
        idx_to_keep = _df[column_name].apply(to_keep)
    if type(to_keep) == type([]):
        idx_to_keep = _df[column_name].apply(lambda x: x in to_keep)

    return _df[idx_to_keep]

def set_figure_size(len_x, len_y):
    sns.set(rc={'figure.figsize':(len_x, len_y)})

def loading_dataset():
    df_data = pd.read_csv('.\hd.csv', index_col=0)
    df_data = df_data[['adj_close', 'volume', 'date']]
    df_data = df_data.rename({'adj_close': 'close'}, axis=1)
    return df_data

def generate_features(df_data, is_improvement=False):
    global N_LAG_REVERSION
    global DECAY_FACTOR_OF_EWMA
```

```

df_data['reversion'] =
df_data['close'].rolling(N_LAG_REVERSION).apply(lambda x: (x[-1] - x.mean()) /
x.std())
    if not is_improvement:
        df_data['momentum'] = df_data['close'] -
df_data['close'].ewm(alpha=DECAY_FACTOR_OF_EWMA,
adjust=True).mean()
#     if is_improvement:
#
#         df_data['momentum'] =
df_data['close'].rolling(DECAY_FACTOR_OF_EWMA).mean().diff(3)
        df_data = df_data.dropna() # df.expanding method requires non-NA
        f_normalize = lambda x: (x[-1] - x.mean()) / (x.max() - x.min())
        df_data.loc[:, ['momentum', 'reversion']] = df_data[['momentum',
'reversion']].expanding().apply(f_normalize)
    return df_data
def generate_trade_position(df_data, is_improvement=False):
    global N_LAG_SIGNAL_DECAY
    if not is_improvement:
        df_data['trade_position'] = df_data['reversion'] + df_data['momentum']
#     if is_improvement:
#
#         df_data['trade_position'] = df_data['reversion'] + df_data['momentum']
        df_data.loc[:, 'trade_position'] =
df_data['trade_position'].rolling(N_LAG_SIGNAL_DECAY).mean()
        # Renormalize trade position to range (-1, 1)
        f_normalize = lambda x: 2 * (x[-1] - x.min()) / (x.max() - x.min()) - 1
        df_data = df_data.dropna() # df.expanding method requires non-NA
        df_data.loc[:, 'trade_position'] =
df_data['trade_position'].expanding().apply(f_normalize)
        df_data.loc[:, 'trade_position'] = df_data['trade_position'].shift(1) # Only use
today's position to predict tomorrow
    return df_data
def get_splited_dataset(df_data):
    df_data = transform_time(df_data, time_col_name='date', form='%Y-%m-%d')

```

```

    in_sample = get_slice(df_data, 'date', lambda dt: dt.year >= 2000 and dt.year <=
2015)

    out_sample = get_slice(df_data, 'date', lambda dt: dt.year >= 2015)
    in_sample = in_sample.set_index('date')
    out_sample = out_sample.set_index('date')
    return in_sample, out_sample
def generate_log_return(df_data):
    df_data['log_return'] = np.log(df_data['close']).diff(1)
    return df_data
def generate_transaction_cost(df_data):
    global UNIT_COST
    df_data['trans_cost'] = UNIT_COST * np.abs(df_data['trade_position'].diff(1))
    return df_data
def get_max_drawdown(daily_profit):
    max_drawdown = 0
    for i in range(len(daily_profit)):
        draw_down = np.max(daily_profit[:i]) - daily_profit[i]
        if draw_down > max_drawdown:
            max_drawdown = draw_down
    return max_drawdown
def get_sharpe(daily_ret, div_std=False):
    if div_std:
        return daily_ret.mean() * 255 / daily_ret.std()
    else:
        return daily_ret.mean() * 255 / get_max_drawdown(daily_ret)
def main_back_test(is_improvement=False):
    """
    Main function of Backtesting.
    Output: In-sample df, Out-sample df
    """
    df_data = loading_dataset()
    df_data = generate_features(df_data, is_improvement)
    df_data = generate_trade_position(df_data, is_improvement)

```

```

df_data = generate_log_return(df_data)
df_data = generate_transaction_cost(df_data)
df_data['profit_pct'] = df_data['log_return'] * df_data['trade_position'] -
df_data['trans_cost'] # 'profit_pct' is profit percentage
df_in_sample, df_out_sample = get_splited_dataset(df_data)
print(get_sharpe(df_in_sample['profit_pct'], div_std=False))
print(get_sharpe(df_in_sample['profit_pct'], div_std=True))
return df_in_sample, df_out_sample

## Time Series plot of Return
get_sharpe(df_in_sample['log_return'])
df_in_sample['close'].plot()
plt.title('Close Price over time', fontsize=20)

## Compare Buy and Hold
UNIT_COST = 0.0005
N_LAG_REVERSION = 5
DECAY_FACTOR_OF_EWMA = 3
N_LAG_SIGNAL_DECAY = 5
df_in_sample, _ = main_back_test(is_improvement=True)
sns.lineplot(data=df_in_sample['log_return'].cumsum(), label='Buy and Hold')
sns.lineplot(data=df_in_sample['profit_pct'].cumsum(), label='Strategy')
plt.title('PnL graph', fontsize=20)

## Parameters Test
#### Lag of Zscore (Reversion)
set_figure_size(13, 5)
UNIT_COST = 0.0001
DECAY_FACTOR_OF_EWMA = 0.96
N_LAG_SIGNAL_DECAY = 5
for N_LAG_REVERSION in [5, 8, 10, 20, 60, 100]:
    df_in_sample, _ = main_back_test()
    sns.lineplot(data=df_in_sample['profit_pct'].cumsum(), label='Window Length
for Z-score: ' + str(N_LAG_REVERSION))
#### Decay Factor of EWMA
set_figure_size(13, 5)

```

```

UNIT_COST = 0.0005
N_LAG_REVERSION = 5
N_LAG_SIGNAL_DECAY = 5
for DECAY_FACTOR_OF_EWMA in [0.05, 0.25, 0.5, 0.75, 0.96, 0.999]:
    df_in_sample, _ = main_back_test()
    sns.lineplot(data=df_in_sample['profit_pct'].cumsum(), label='Decay factor for
EWMA: ' + str(DECAY_FACTOR_OF_EWMA))
### Signal Decay
UNIT_COST = 0.0005
N_LAG_REVERSION = 5
DECAY_FACTOR_OF_EWMA = 0.96
for N_LAG_SIGNAL_DECAY in [2, 5, 10, 20, 40]:
    df_in_sample, _ = main_back_test()
    sns.lineplot(data=df_in_sample['profit_pct'].cumsum(), label='Lag of Signal
Decay: ' + str(N_LAG_SIGNAL_DECAY))
## Optimal Parameter comparing to BAH
UNIT_COST = 0.0005
N_LAG_REVERSION = 5
DECAY_FACTOR_OF_EWMA = 0.96
N_LAG_SIGNAL_DECAY = 5
df_in_sample, _ = main_back_test()
sns.lineplot(data=df_in_sample['log_return'].cumsum(), label='Buy and Hold')
sns.lineplot(data=df_in_sample['profit_pct'].cumsum(), label='Strategy')
plt.title('PnL graph', fontsize=20)
## Out-sample Performance
UNIT_COST = 0.0005
N_LAG_REVERSION = 5
DECAY_FACTOR_OF_EWMA = 0.96
N_LAG_SIGNAL_DECAY = 5
_, df_out_sample = main_back_test()
sns.lineplot(data=df_out_sample['log_return'].cumsum(), label='Buy and Hold')
sns.lineplot(data=df_out_sample['profit_pct'].cumsum(), label='Strategy')
plt.title('PnL graph', fontsize=20)

```

```

## Improvement
UNIT_COST = 0.0001
N_LAG_REVERSION = 5
DECAY_FACTOR_OF_EWMA = 2
N_LAG_SIGNAL_DECAY = 5
_, df_in_sample = main_back_test(is_improvement=True)
sns.lineplot(data=df_in_sample['log_return'].cumsum(), label='Buy and Hold')
sns.lineplot(data=df_in_sample['profit_pct'].cumsum(), label='Strategy')
plt.title('PnL graph', fontsize=20)
set_figure_size(13, 5)

UNIT_COST = 0.0005
N_LAG_REVERSION = 5
N_LAG_SIGNAL_DECAY = 5

for DECAY_FACTOR_OF_EWMA in [2, 3, 5, 10, 20]:
    df_in_sample, _ = main_back_test(is_improvement=True)
    sns.lineplot(data=df_in_sample['profit_pct'].cumsum(), label='Decay factor for
EWMA: ' + str(DECAY_FACTOR_OF_EWMA))

```